DTU

Model-based Machine Learning

Temporal models

Outline

Case study: Modeling freeway occupancy rates in San Francisco

Autoregressive models

State-space models

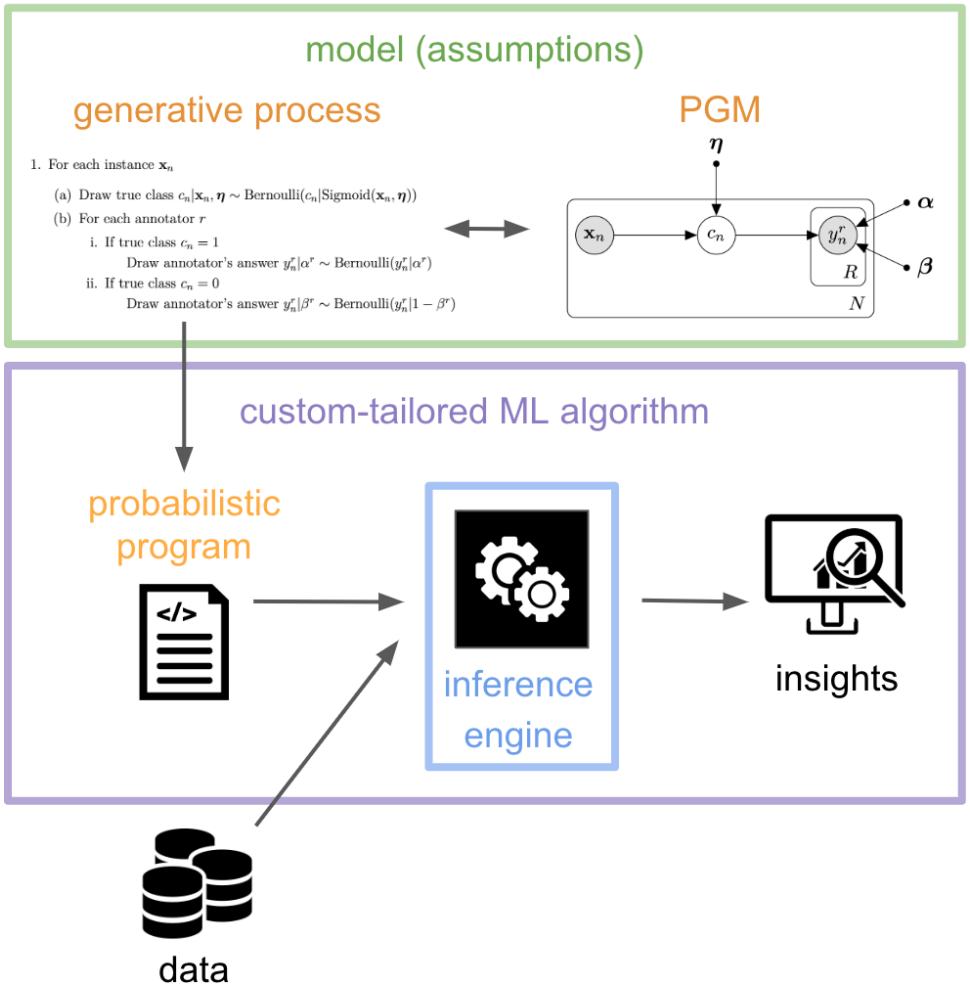
Multivariate state-space models

Road to MBML: where are we?

Foundations of PGMs and probabilistic programming
Probabilistic modelling in various contexts - your modelling toolbox
Bayesian inference (exact and approximate)

Week	Topic
1	Intro to the course + Prob. review
2	PGM foundations
3	PGM foundations II
4	Freq. vs Bayesian + Prob. Prog. + Mixture models
5	Regression models
6	Classification and Hierarchical models
7	Temporal models
8	Topic Models
9	Markov-chain Monte Carlo (MCMC)
10	Variational inference
11	Generative models
12	Gaussian processes
13	Project support

Model-based Machine Learning



Learning objectives

At the end of this lecture, you should be able to:

- Explain what Markov models are and their underlying assumptions
- Explain the underlying concepts and assumptions behind state-space models
- Explain the difference between forecasting and imputation
- Relate different ways of modelling the temporal dependency between variables and justify their suitability for a problem
- Implement different temporal modelling techniques in Pyro

Modeling freeway occupancy rates in San Francisco

- Sensors measure occupancy rate (between 0 and 1) of different car lanes
- Time-series of measurements every 10 minutes (144 observations per day)
- Multivariate time-series data: one time-series per sensor

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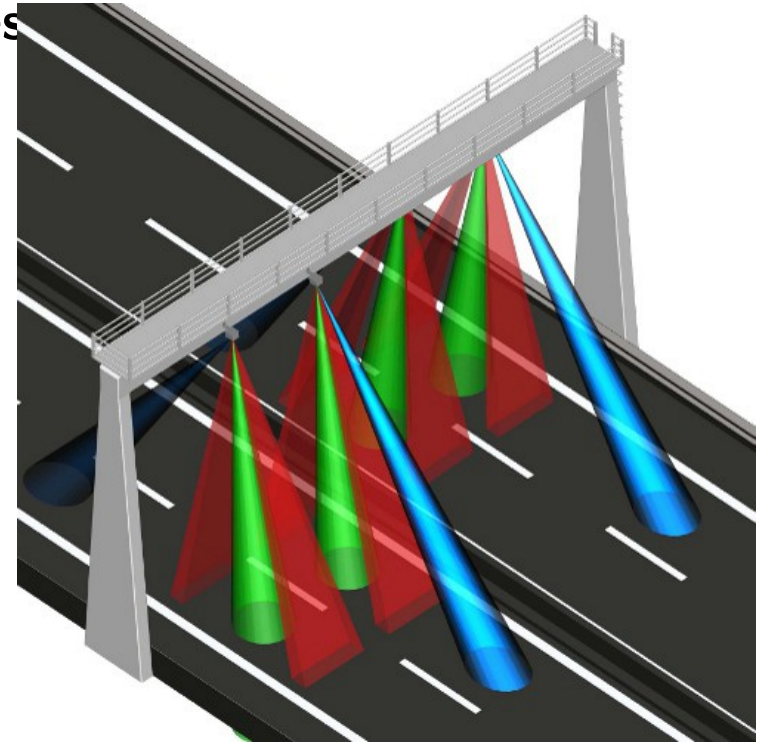
- Goal: **model freeway occupancy rates**
- Some possible applications:
  - Predict future occupancy rates for better routing
  - Identify problems and send alerts to road users
  - Understand drivers' behaviours



Modeling freeway occupancy rates in San Francisco (cont'

- Let's start thinking about the graphical model...
- We shall start by considering **only one sensor**
- Let y_t denote the occupancy rate in that sensor at time t

$$y_1 \qquad y_2 \qquad y_3 \qquad \dots \qquad y_T$$

- How should we connect these (dependencies)?

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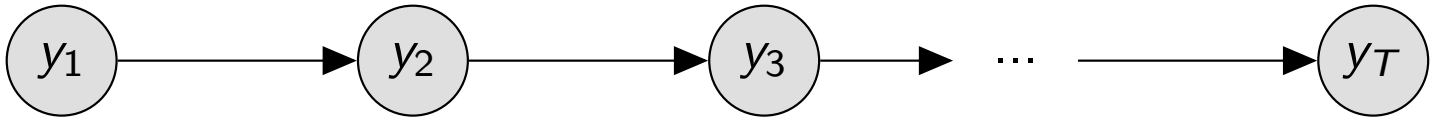

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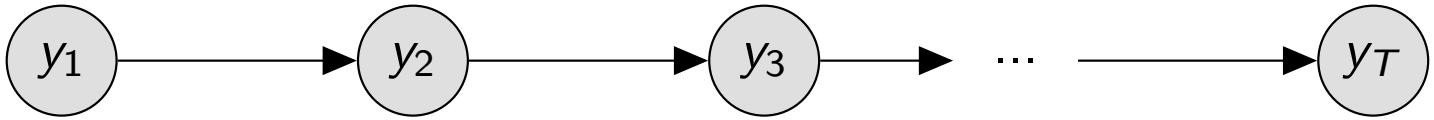

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- What other variables could we have considered?
  - Seasonal information
  - Day of the week
  - Weather data
  - Information from other sensors
  - ...but for now we shall consider only data from the time-series $\{y_1, \ldots, y_T\}$

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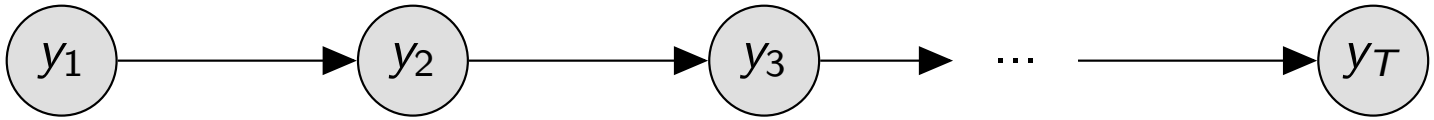

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- How should we model the dependency of y_t on y_{t-1}?

Markov models (or Markov chains)

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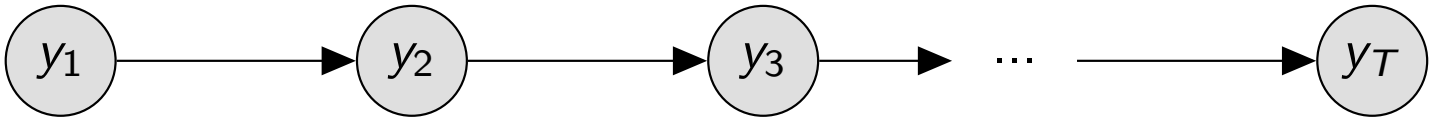

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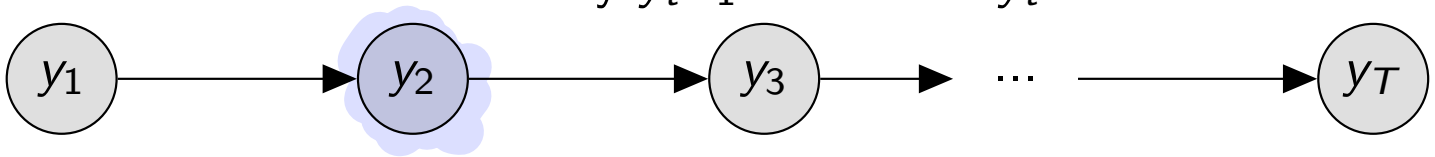

- Joint distribution of all variables factorizes as

$$\boxed{p(y_1, y_2, \ldots, y_T) = p(y_1) \prod_{t=2}^{T} p(y_t | y_{t-1})}$$

always the same over time

- This only requires us to specify a **transition probability** $p(y_t | y_{t-1})$ (well, and some prior distribution for the first observation y_1)

Note

Transition probabilities are time-independent (same transition distribution for all t)

Autoregressive (AR) models

- Standard and widely used Markov model of continuous observations
- Assume that each y_t is a **linear function** of the M previous observations (plus some Gaussian-distributed observation noise term $\epsilon \sim \mathcal{N}(0, \sigma^2)$)

$$y_t = \beta_1 y_{t-1} + \beta_2 y_{t-2} + \cdots + \beta_M y_{t-M} + \epsilon$$

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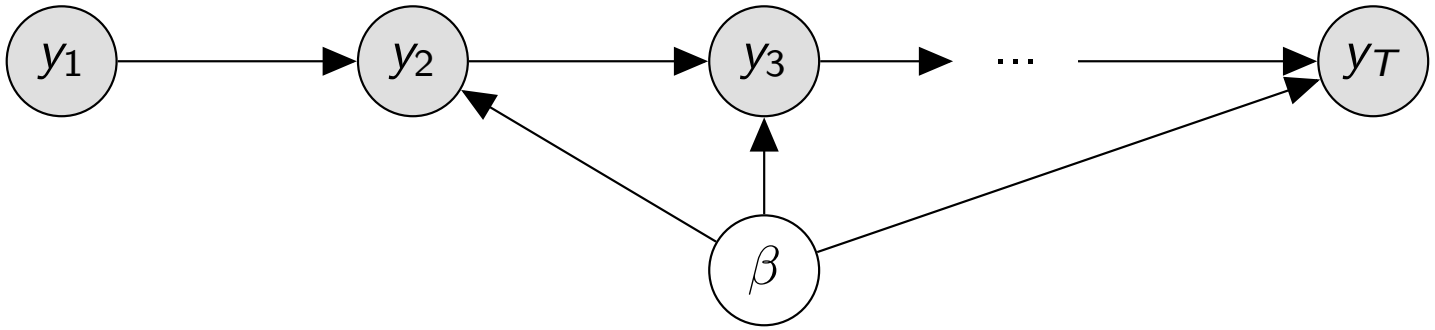
- Graphical model for (Bayesian) autoregressive model of order 1 (AR1)



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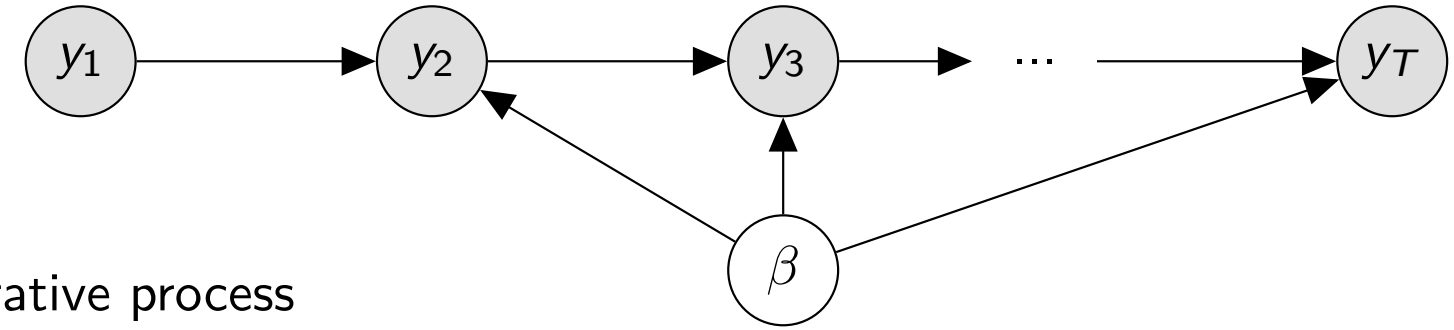
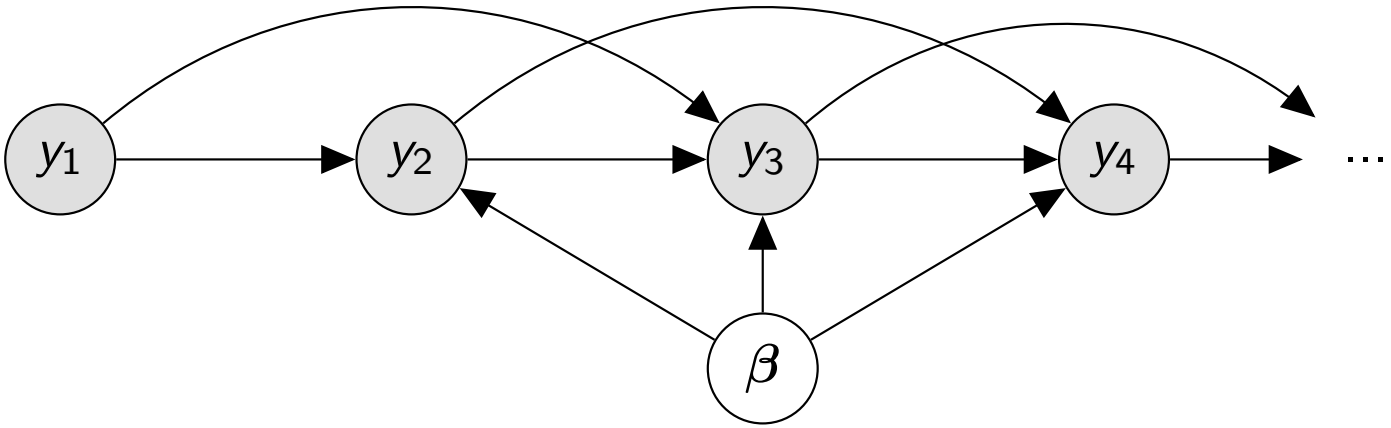


- Generative process
  1. Draw transition coefficient $\beta \sim \mathcal{N}(\beta|0, \lambda)$
  2. Draw first observation $y_1 \sim \mathcal{N}(y_1|\mu_0, \sigma_0^2)$ doesn't depend on anything
  3. For each time $t \in \{2, \ldots, T\}$
     a. Draw observation $y_t \sim \mathcal{N}(y_t|\beta y_{t-1}, \sigma^2)$

Autoregressive (AR) models: higher-level dependencies

- Graphical model for (Bayesian) autoregressive model of order 2 (AR2)



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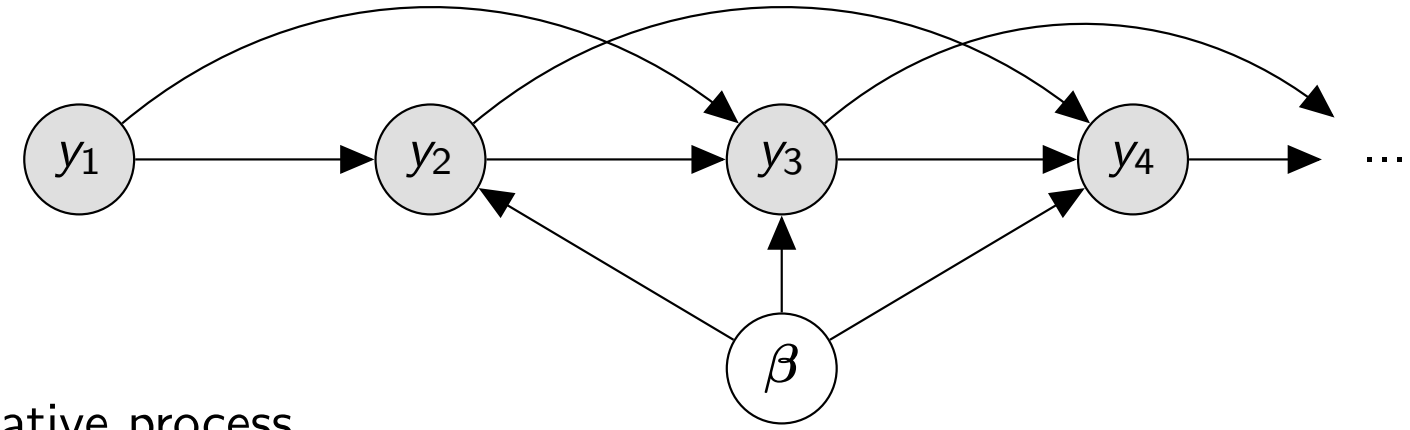

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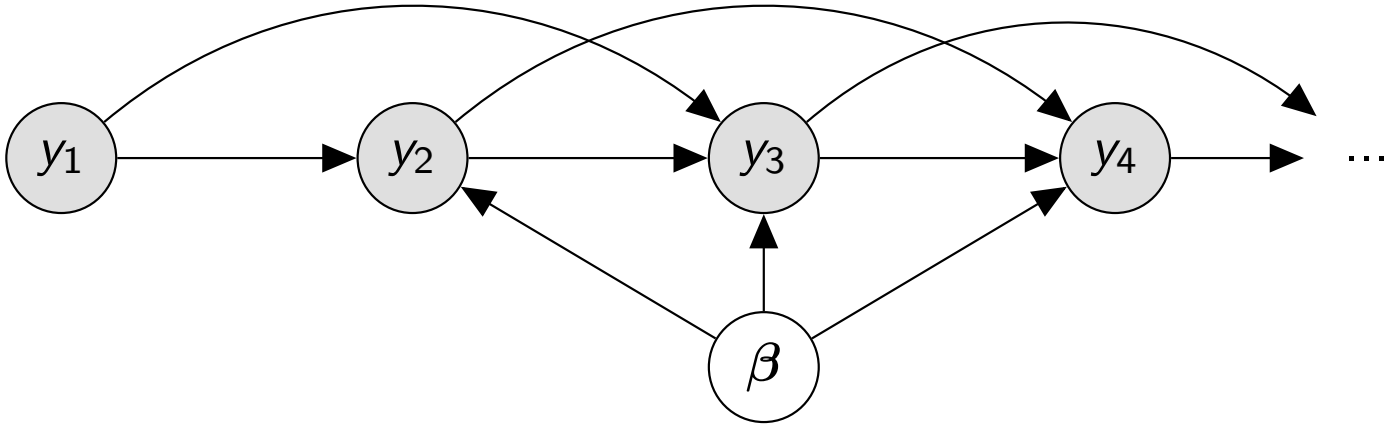
  3. Draw second observation $y_2 \sim \mathcal{N}(y_2|\beta_1 y_{t-1}, \sigma^2)$
  4. For each time $t \in \{3, \ldots, T\}$
     a. Draw observation $y_t \sim \mathcal{N}(y_t|\beta_1 y_{t-1} + \beta_2 y_{t-2}, \sigma^2)$

Autoregressive (AR) models: summary so far...



- In autoregressive models, each observation y_t is a linear function of the M previous observations $\{y_{t-1}, \ldots, y_{t-M}\}$ plus some noise
- But, if $\{y_{t-1}, \ldots, y_{t-M}\}$ are themselves noisy observations (e.g. due to measurement noise in sensors), aren't we building up noise over time?
- Perhaps there is a better way to model temporal data...

State-space models (SSMs)

- In AR models, only the observations $\{y_1, \ldots, y_T\}$ are modeled explicitly
- More general models of time-series introduce an extra set of unobserved (hidden) variables $\{\mathbf{h}_1, \ldots, \mathbf{h}_T\}$ from which the observations are assumed to be generated

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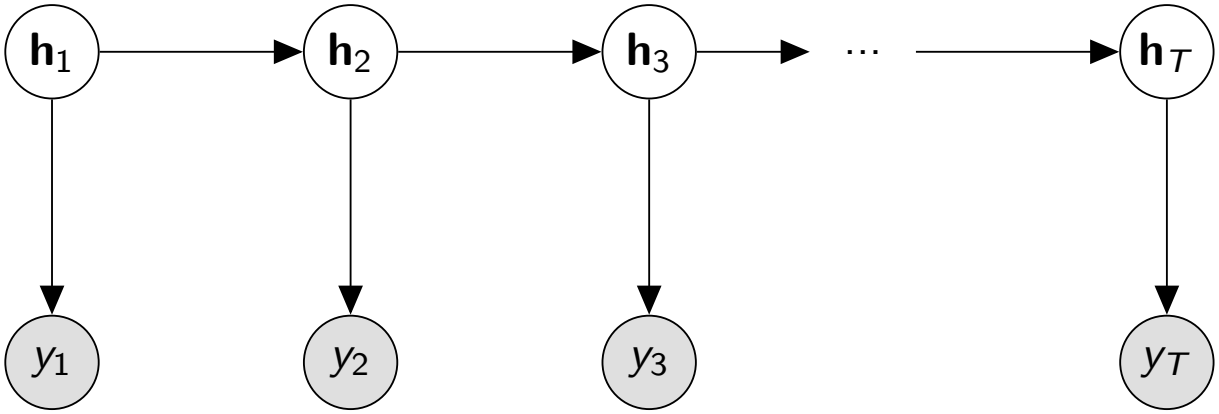
- y_t is assumed to be independent from all other variables given $\mathbf{h}_t$



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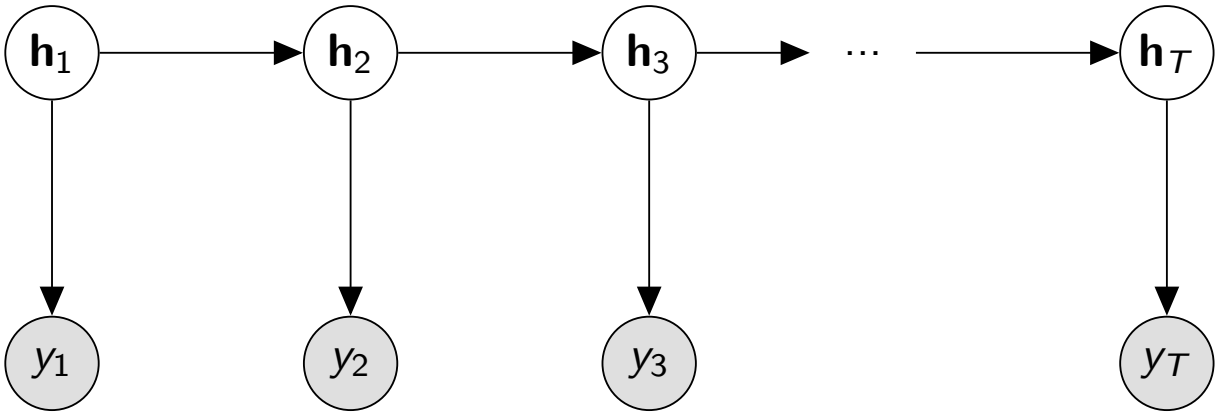

- Notice that the latent **states** $\mathbf{h}_t$ can be vector-valued!
- Many physical processes can be expressed in this state-space framework

Linear dynamical systems (LDSs)

- State-space models require us to specify two probability distributions:
  - Transition probability: $p(\mathbf{h}_t|\mathbf{h}_{t-1})$
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  - Efficient (exact!) inference of latent states $\mathbf{h}_t$ (the Kalman filter)

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governs

covariance matrix

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maps from hidden state to actual variables

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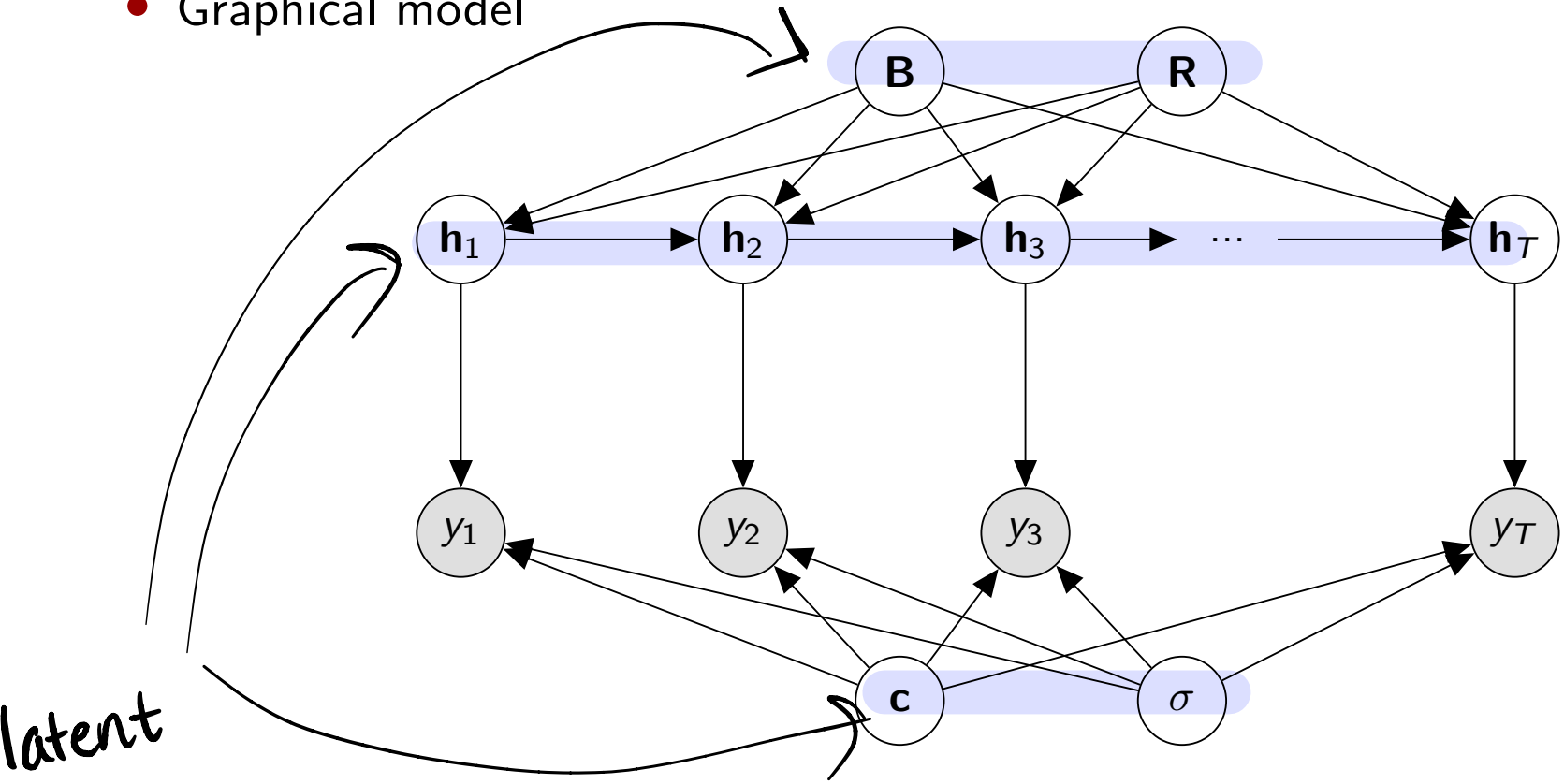
- We shall consider a fully Bayesian approach
  - Assign priors to $\mathbf{B}$, $\mathbf{R}$, $\mathbf{c}$ and σ
  - Use Bayesian inference to compute respective posteriors

Linear dynamical systems (LDSs)

- Graphical model



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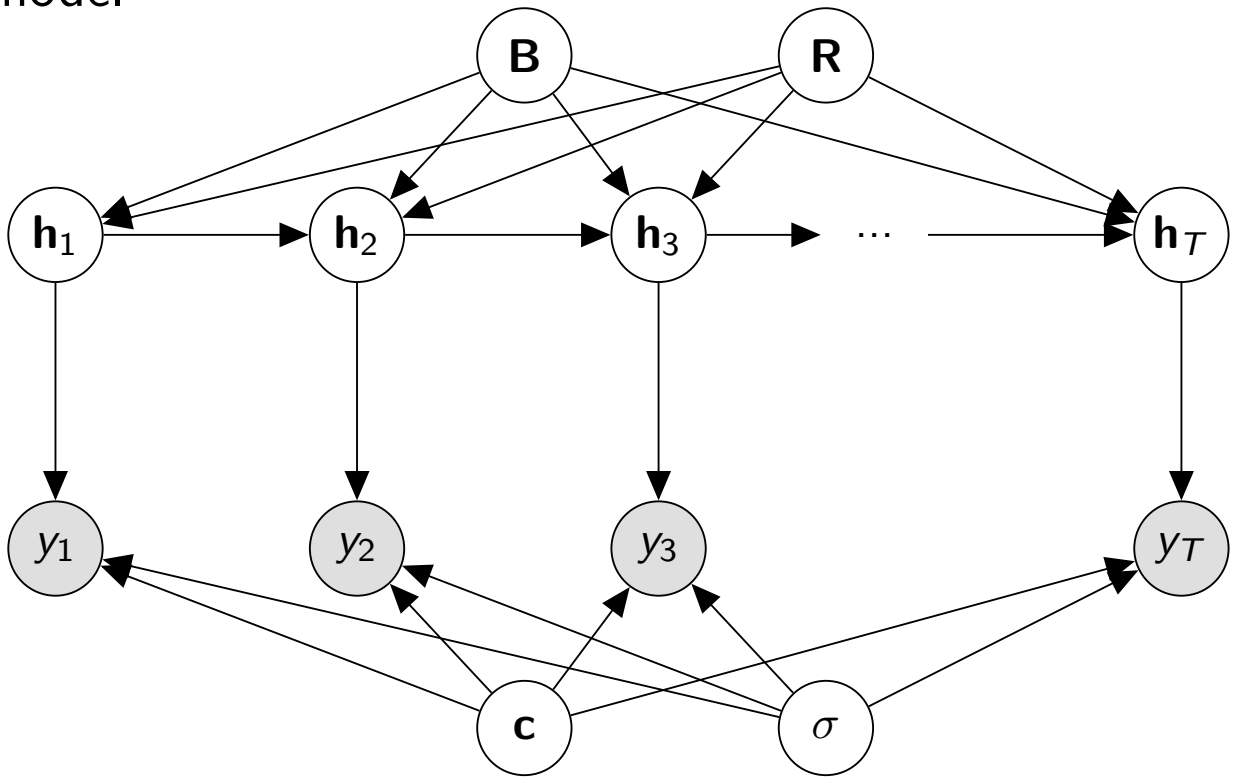


- The goal of inference is to compute the posterior distribution over all the latent variables in the model: $\{\mathbf{h}_1, \ldots, \mathbf{h}_T\}$, $\mathbf{B}$, $\mathbf{R}$, $\mathbf{c}$ and σ
  - Can be very challenging (e.g. identifiability issues)!
  - We often need to impose some restrictions on the structure of $\mathbf{B}$, $\mathbf{R}$, $\mathbf{c}$.

AR models as linear dynamical systems (LDSs)

- By appropriately defining the latent states, many popular time-series models can be cast as an LDS
- E.g. an autoregressive model of order k (AR-k) can be written as:

$$\underbrace{\begin{pmatrix} h_t \\ h_{t-1} \\ \vdots \\ h_{t-k+1} \end{pmatrix}}_{\mathbf{h}_t} = \underbrace{\begin{pmatrix} \beta_1 & \beta_2 & \cdots & \beta_k \\ 1 & 0 & 0 & 0 \\ 0 & \ddots & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}}_{\mathbf{B}} \underbrace{\begin{pmatrix} h_{t-1} \\ h_{t-2} \\ \vdots \\ h_{t-k} \end{pmatrix}}_{\mathbf{h}_{t-1}} + \begin{pmatrix} \tau_t \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

exercise calculate

with the emission probability defined as

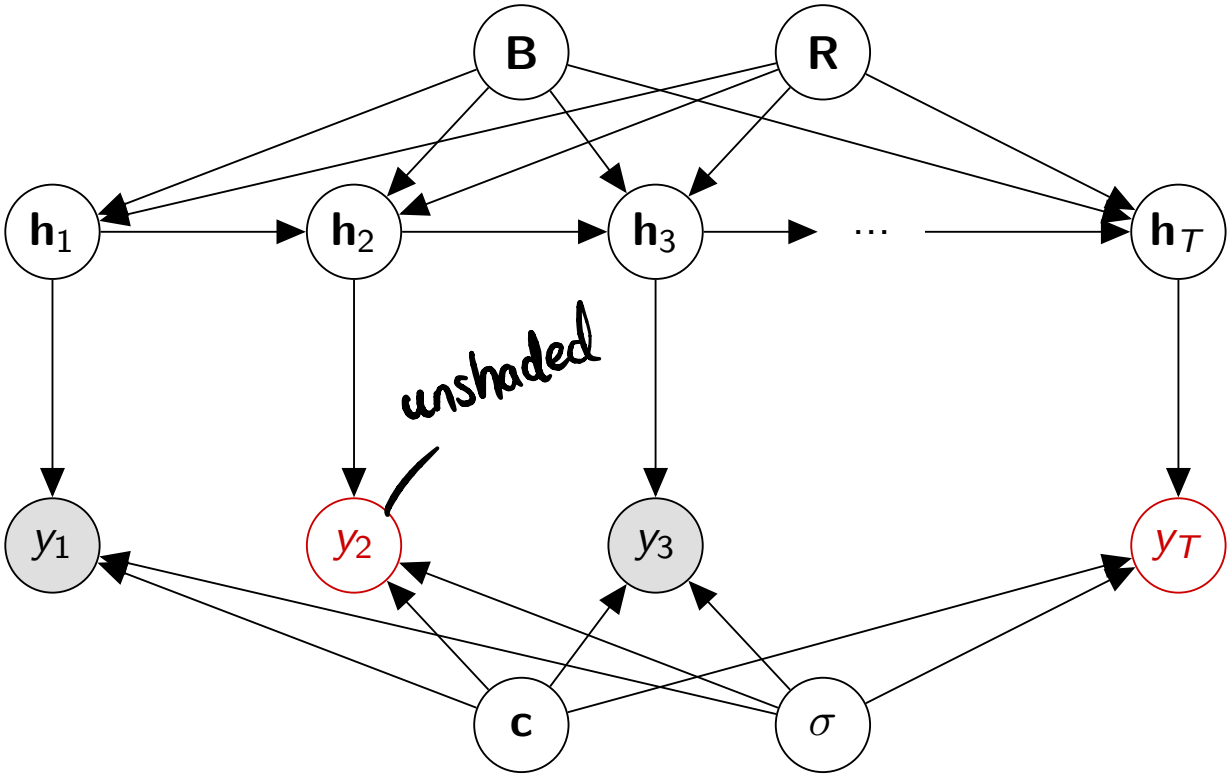
$$y_t = \underbrace{\begin{pmatrix} 1 & 0 & \cdots & 0 \end{pmatrix}}_{\mathbf{c}} \mathbf{h}_t + \epsilon_t$$

where $\tau_t \sim \mathcal{N}(0, r)$ and $\epsilon_t \sim \mathcal{N}(0, \sigma^2)$

- Notice the restrictions that we imposed on the structures of **B** and **c**

Missing data

- What if we have **missing** observations?



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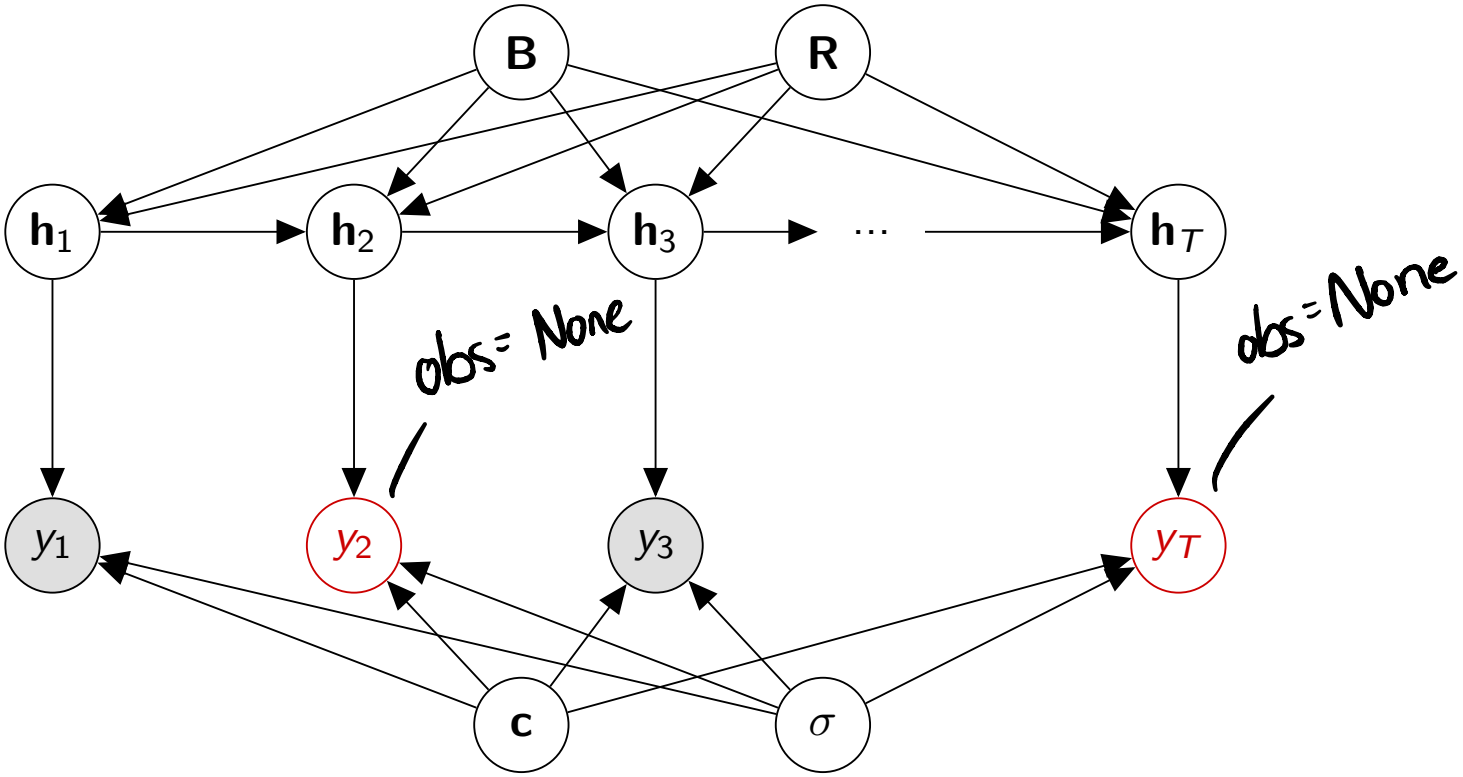
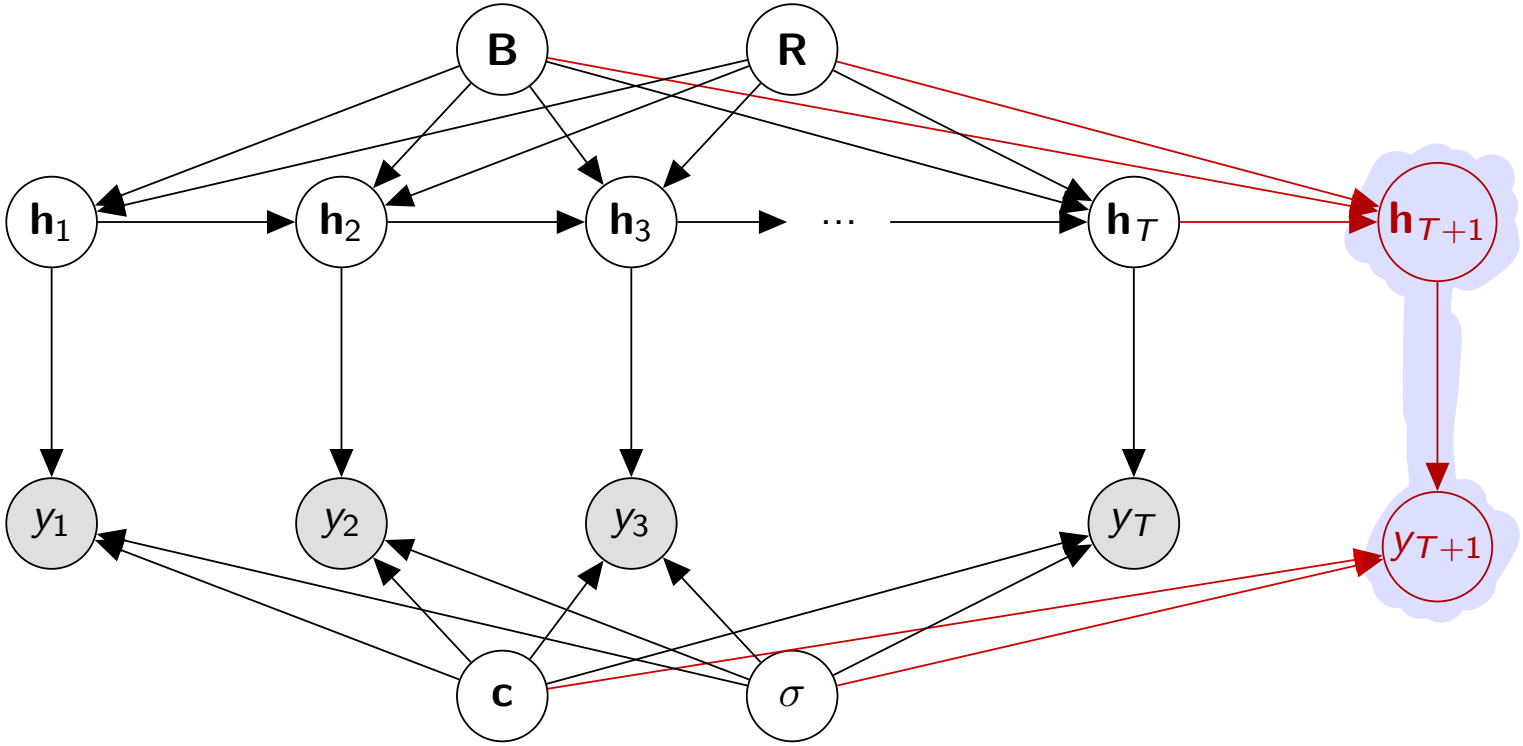


- Just treat the missing observations as latent variables
  - Marginalize out their values
  - Compute their posterior distribution

pyro will do inference on them and will give the posterior

Forecasting

- What if we want to predict the value of y_{T+1}?



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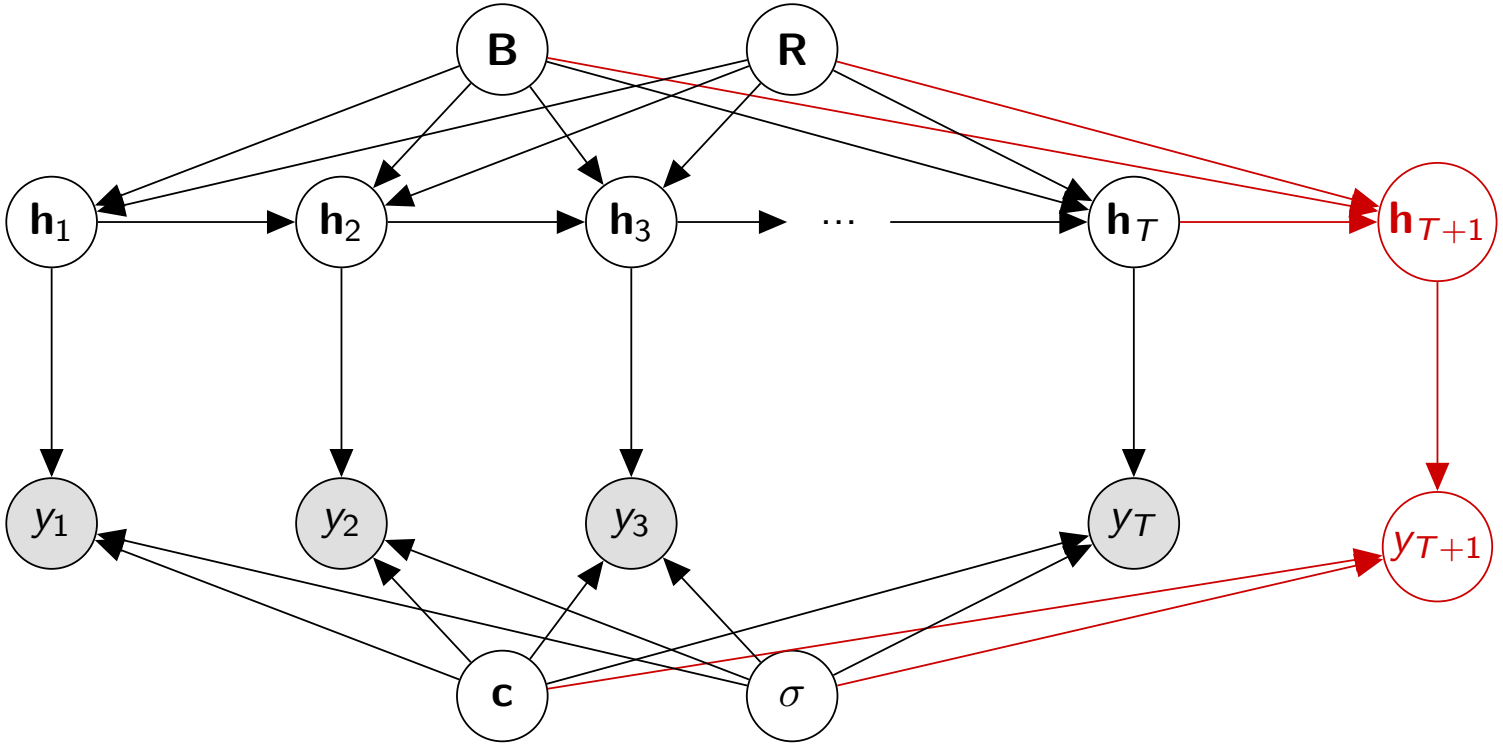
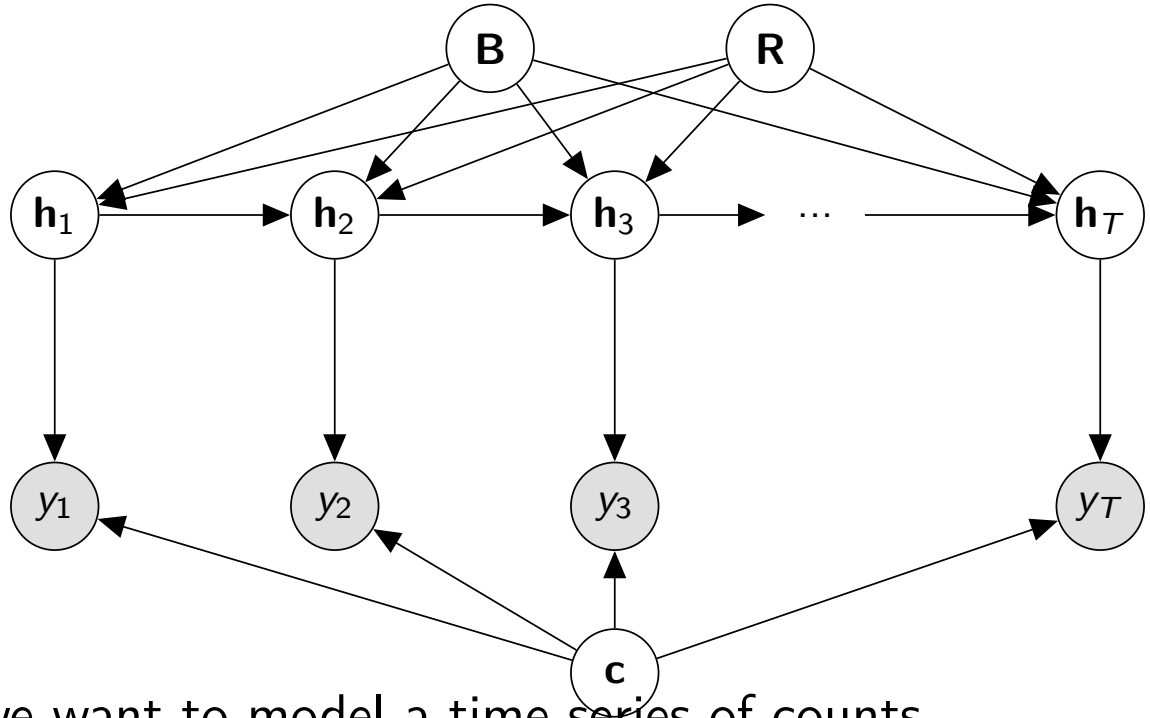


- **Extend** the model for one (or more) time-steps with latent variables
  - Compute the posterior distribution over y_{T+1}

Playtime!

- Ancestral sampling from Bayesian LDS model
  - See “06 - Temporal models - Part 1.ipynb" notebook
  - Expected duration: 10 minutes
- Bayesian LDS model of freeway occupancy rates: forecasting
  - See “06 - Temporal models - Part 2.ipynb" notebook
  - Expected duration: 45 minutes

Extensions: Poisson LDSs



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  - E.g. forecasting the number of arrivals/departures at a metro station

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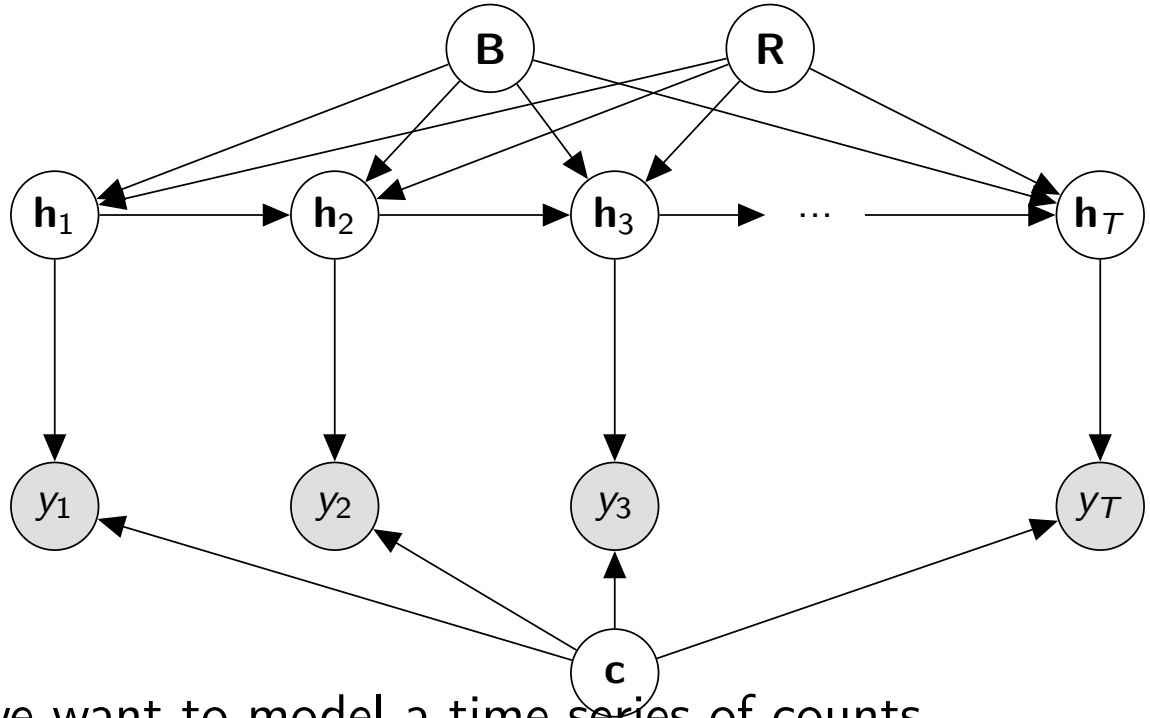

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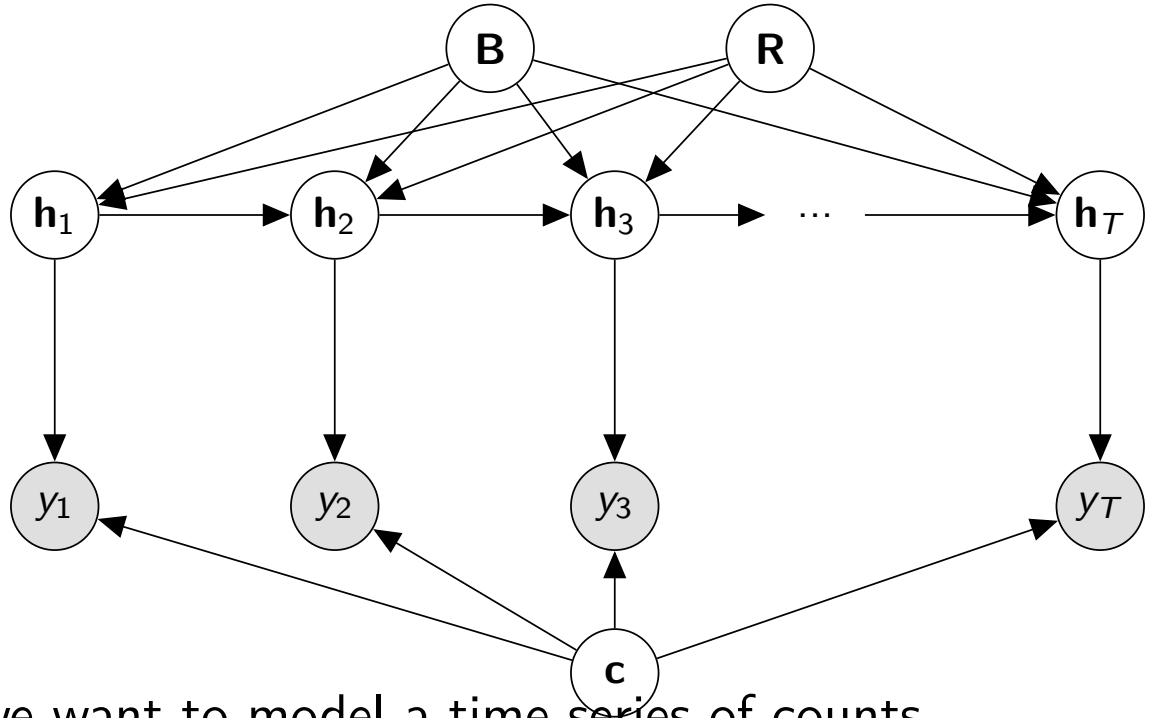

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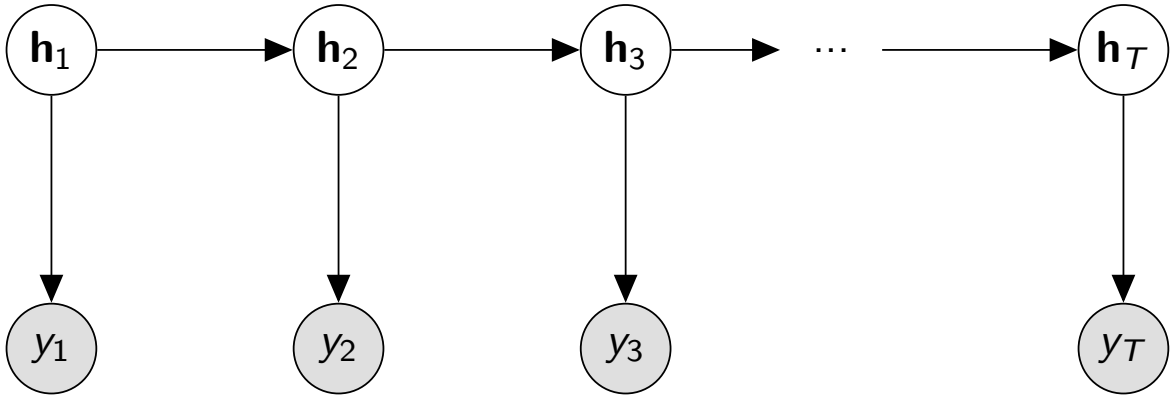
- In fact, as with generalized linear models (GLMs), we could use **any** general

Extensions: Linear dynamical systems (LDSs) with inputs

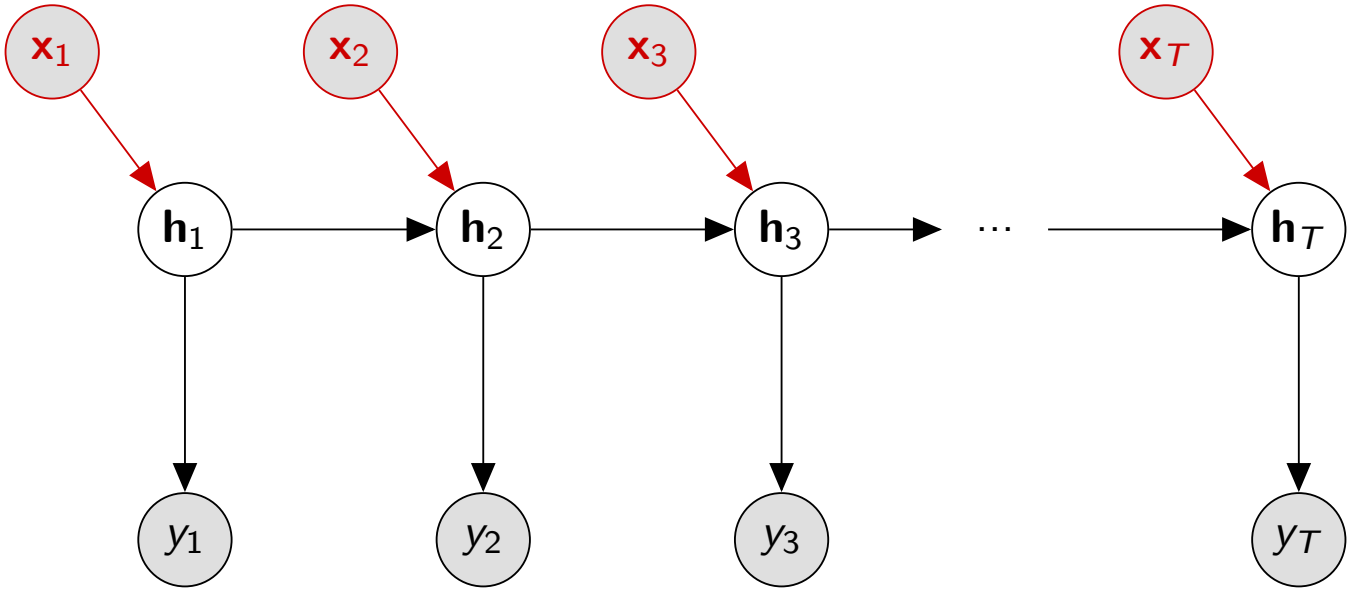
- Graphical model representation of a linear dynamical system:



  (for simplicity, we omitted the variables $\mathbf{B}$, $\mathbf{R}$, $\mathbf{c}$ and σ from the graph)

- What if we have additional input information available at each time t that can help explain the observations y_t?
  - In our case study of freeway occupancy rates, this could be event information, weather, seasonal information, etc.

Extensions: Linear dynamical systems (LDSs) with inputs



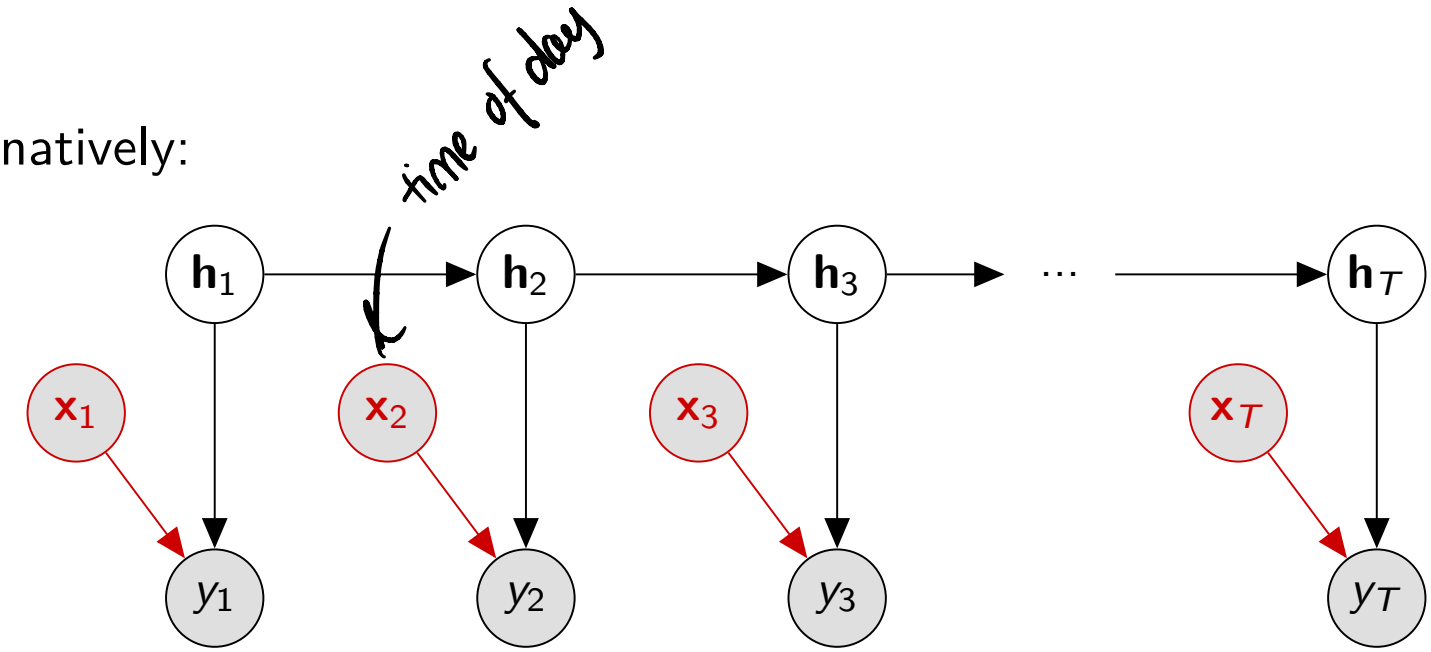
(for simplicity, we omitted the variables $\mathbf{B}$, $\mathbf{R}$, $\mathbf{c}$, σ and $\mathbf{W}$ from the graph)

- $\mathbf{x}_t$ are **external inputs** at time t
- Transition probability becomes:

$$\mathbf{h}_t \sim \mathcal{N}(\mathbf{h}_t | \mathbf{B}\mathbf{h}_{t-1} + \mathbf{W}\mathbf{x}_t, \mathbf{R})$$

Extensions: Linear dynamical systems (LDSs) with inputs

Or, alternatively:



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- $\mathbf{x}_t$ are **external inputs** at time t
- Emission probability becomes:

$$y_t \sim \mathcal{N}(y_t | \mathbf{c}^\mathsf{T}\mathbf{h}_t + \mathbf{W}\mathbf{x}_t, \sigma^2)$$

- What are the implications? How do they differ from the previous slide?

Extensions: Extended Kalman filter (EKF)

- So far we have assumed linear relations between states and from state to observation

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- The *extended Kalman filter* (EKF) relaxes this assumption by allowing these relations to be defined by arbitrary **differentiable functions** f and g

$$\mathbf{h}_t \sim \mathcal{N}(\mathbf{h}_t | f(\mathbf{h}_{t-1}, \mathbf{x}_t), \mathbf{R})$$
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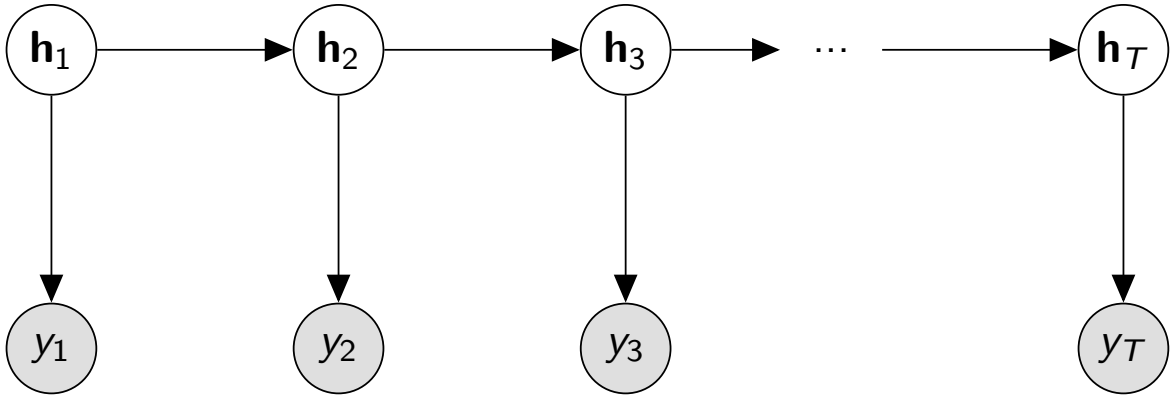
- f and g can even be neural networks! (search for "Deep K)
- EKF allows for a **non-linear** version of the Kalman filter
- Extremely popular in navigation systems and GPS (e.g. your phone uses it all the time!)
- Specialized algorithms have been developed for learning and performing inference in EKFs - not the focus of this course! We will keep relying on Pyro

Extensions: Different regimes

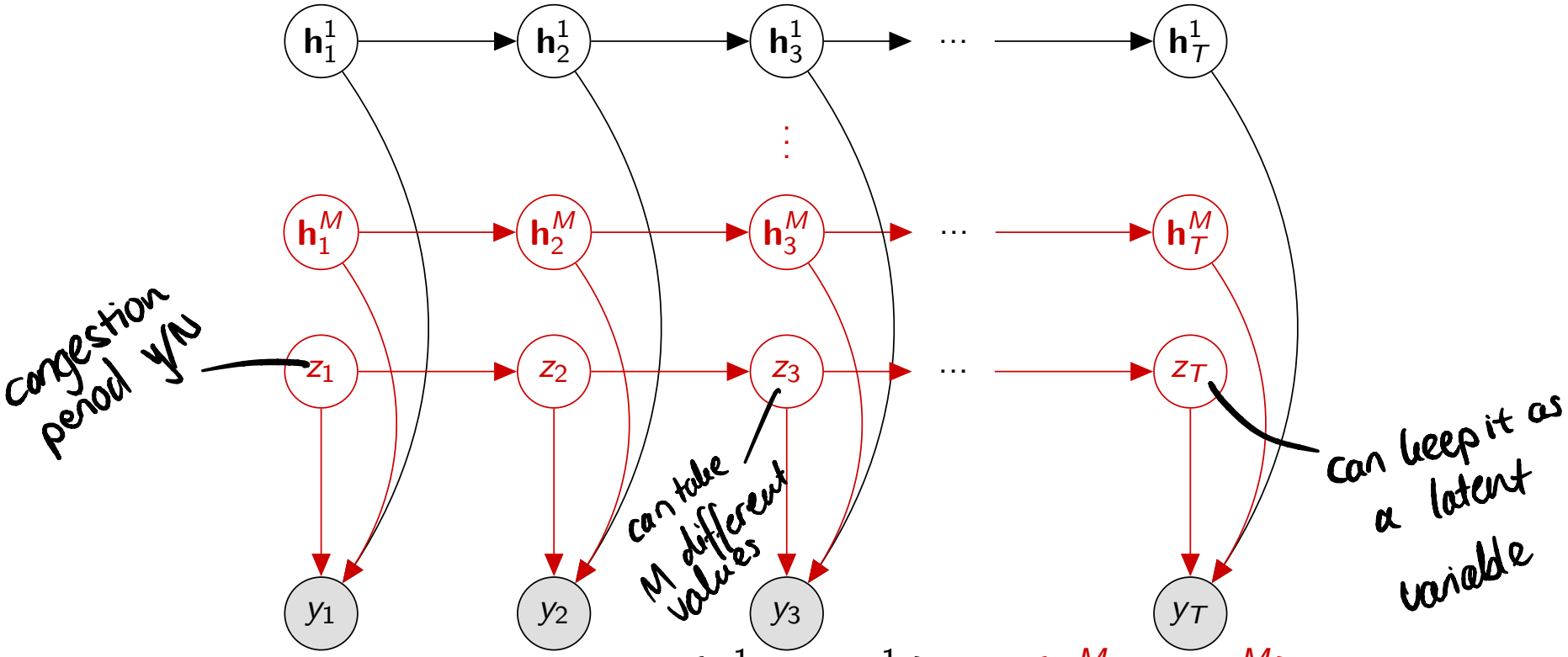
- Graphical model representation of a linear dynamical system:



  (for simplicity, we omitted the variables $\mathbf{B}$, $\mathbf{R}$, $\mathbf{c}$ and σ from the graph)

- Suppose that the time-series data can behave quite differently in different periods (**regimes**)
  - E.g. do traffic conditions evolve the same way over time in congestion periods as they do in free flow?
- How can we change the model to incorporate this prior knowledge?

Extensions: Switching linear dynamical systems (SLDSs)



- Multiple latent state chains: $\{\mathbf{h}_1^1, \ldots, \mathbf{h}_T^1\}$ ... $\{\mathbf{h}_1^M, \ldots, \mathbf{h}_T^M\}$
- At each time t, a discrete latent variable z_t selects one of the latent state chains to be "active" and produce an observation y_t

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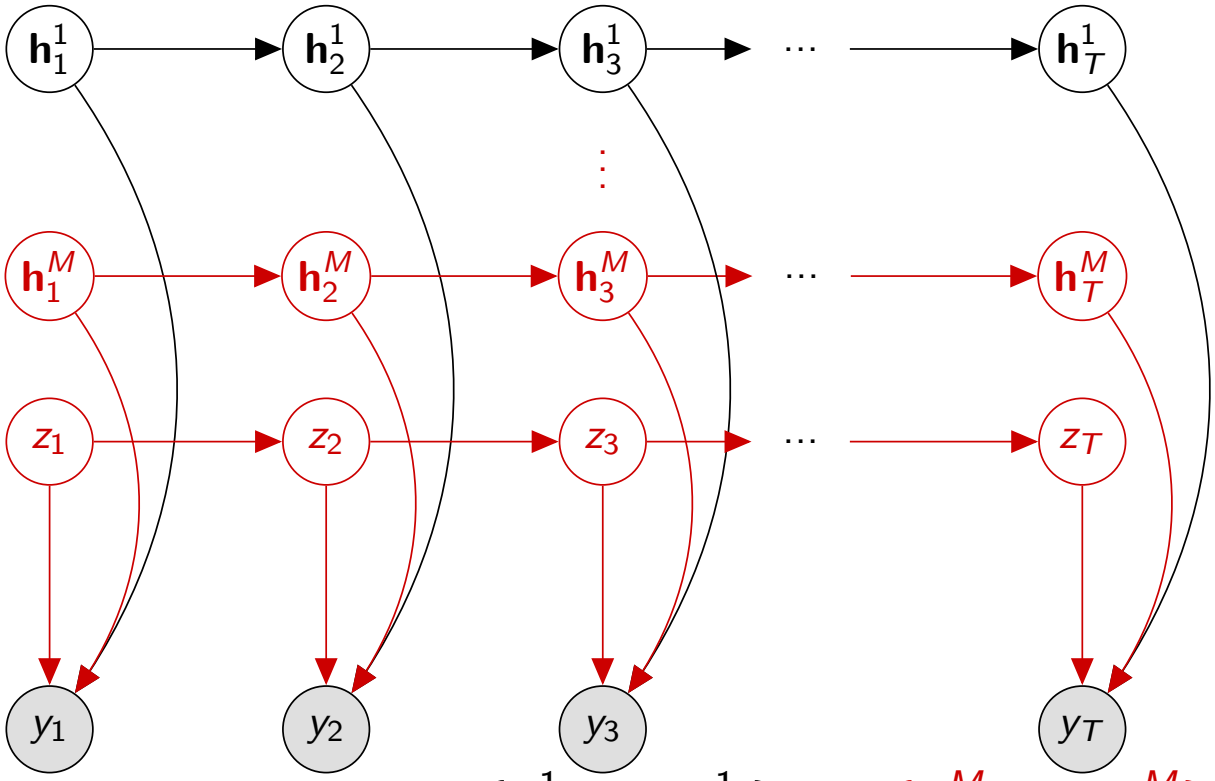
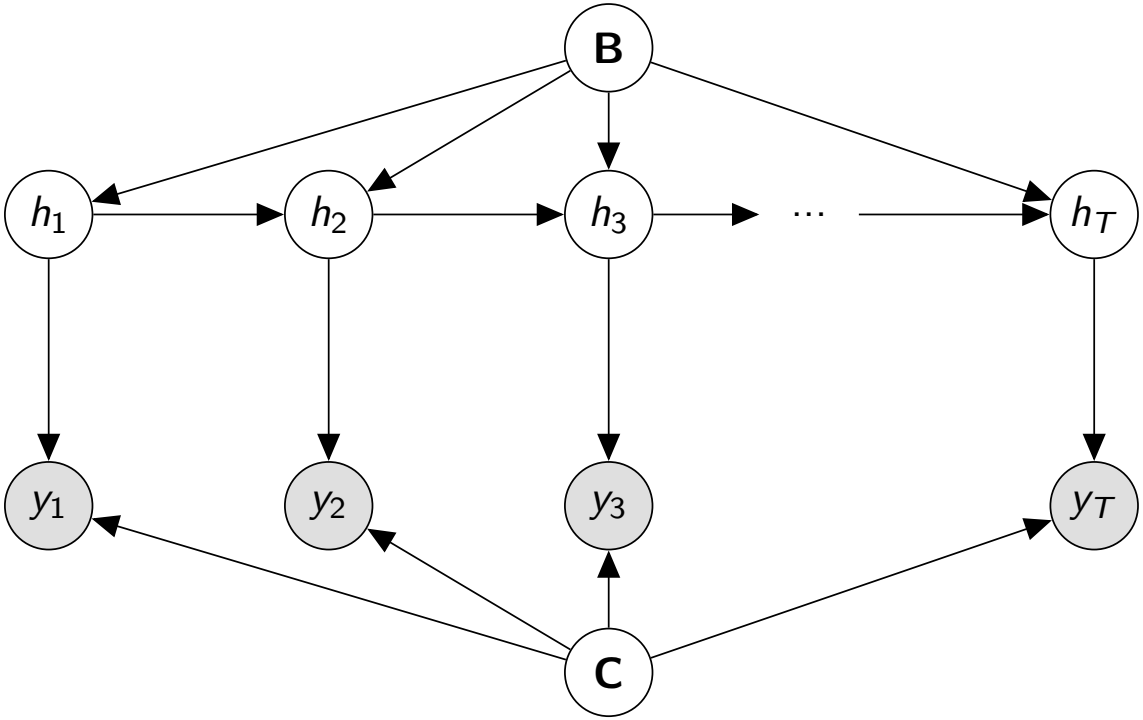


- Multiple latent state chains: $\{\mathbf{h}_1^1, \ldots, \mathbf{h}_T^1\} \ \ldots \ \{\mathbf{h}_1^M, \ldots, \mathbf{h}_T^M\}$
- At each time t, a discrete latent variable z_t selects one of the latent state chains to be "active" and produce an observation y_t
- Useful if your observations can alternate between regimes with different dynamics
  - E.g. road segments with multiple traffic regimes (congested vs. free flow)

Hidden Markov models (HMMs)

- Discrete analog of the LDS is the *hidden Markov model* (HMM)



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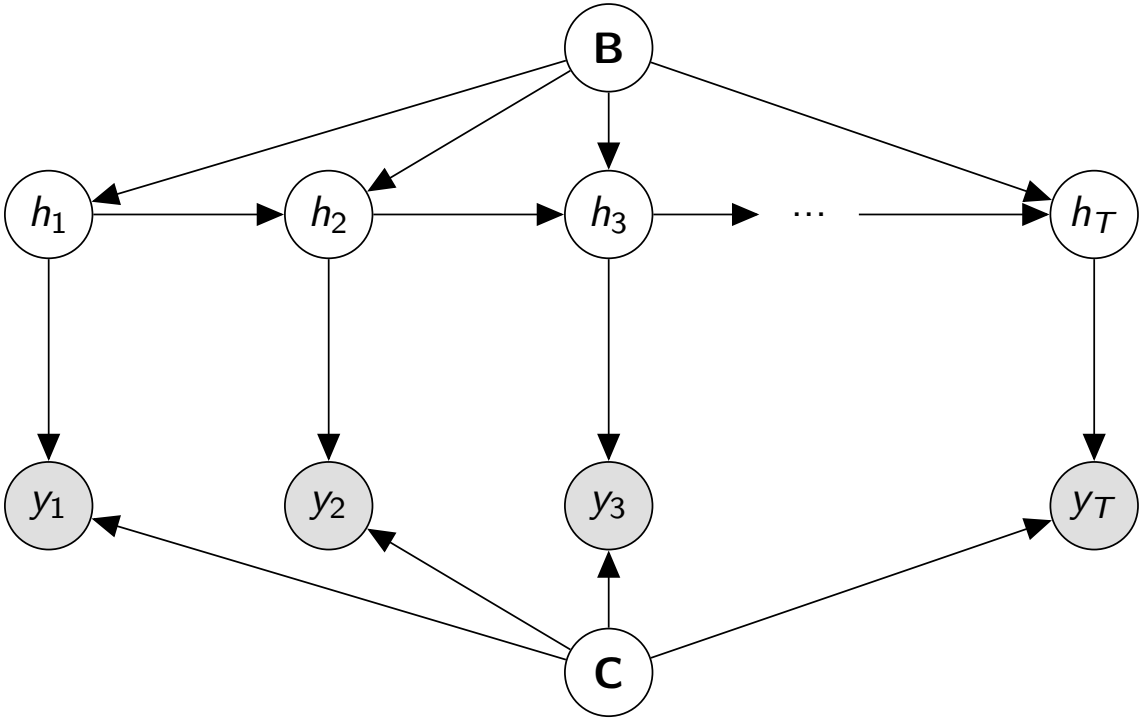

- Hidden states h_t and observations y_t are **discrete random variables**
- Transition probabilities $p(h_t|h_{t-1}, \mathbf{B})$ and emission probabilities $p(y_t|h_t, \mathbf{C})$ can be simply represented using conditional probability tables (CPTs)
  - Parameter matrices **B** and **C** are CPTs!

Hidden Markov models (HMMs)

- Suppose that the latent state h_t can take K possible values: $h_t \in \{1, \ldots, K\}$
- We can specify the transition probabilities using a $K \times K$ matrix $\mathbf{B}$
  - Element b_{ij} of matrix $\mathbf{B}$ denotes the probability of transition to state j given that the current state is i
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  - Element c_{ij} of matrix $\mathbf{C}$ denotes the probability of observing the value $y_t = j$ given that the current state h_t is i
  - Given that the current state h_t is i, the row $\mathbf{c}_i$ specifies the probabilities of observing each of the possible V values at time t ($y_t \in \{1, \ldots, V\}$)

- Notice that the rows $\mathbf{b}_i$ and $\mathbf{c}_i$ must sum to 1 in order to be valid probabilities!

$$\beta = \begin{array}{cc} & h_t \\ \begin{bmatrix} 0.1 & 0.3 & 0.6 \\ 0.7 & 0.1 & 0.2 \\ 0.8 & 0.1 & 0.1 \end{bmatrix} & h_{t-1} \\ K \times K & \end{array}$$

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$$h_t \sim \text{Categorical}(h_t | \mathbf{b}_{h_{t-1}})$$

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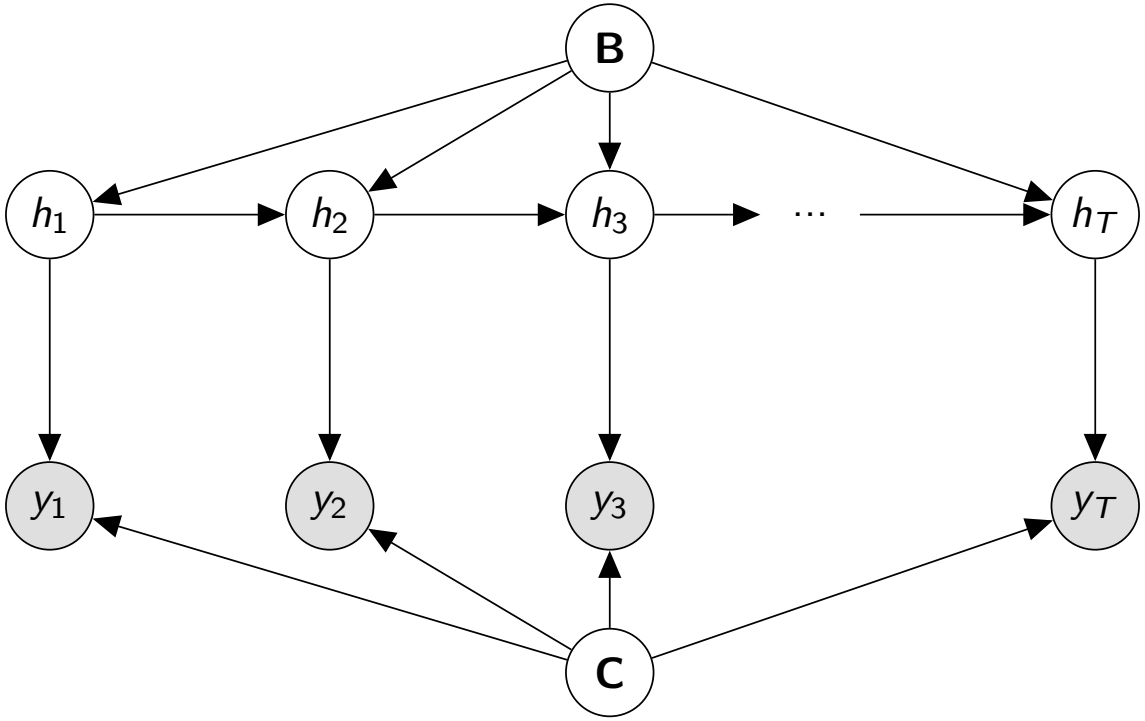
Hidden Markov models (HMMs)



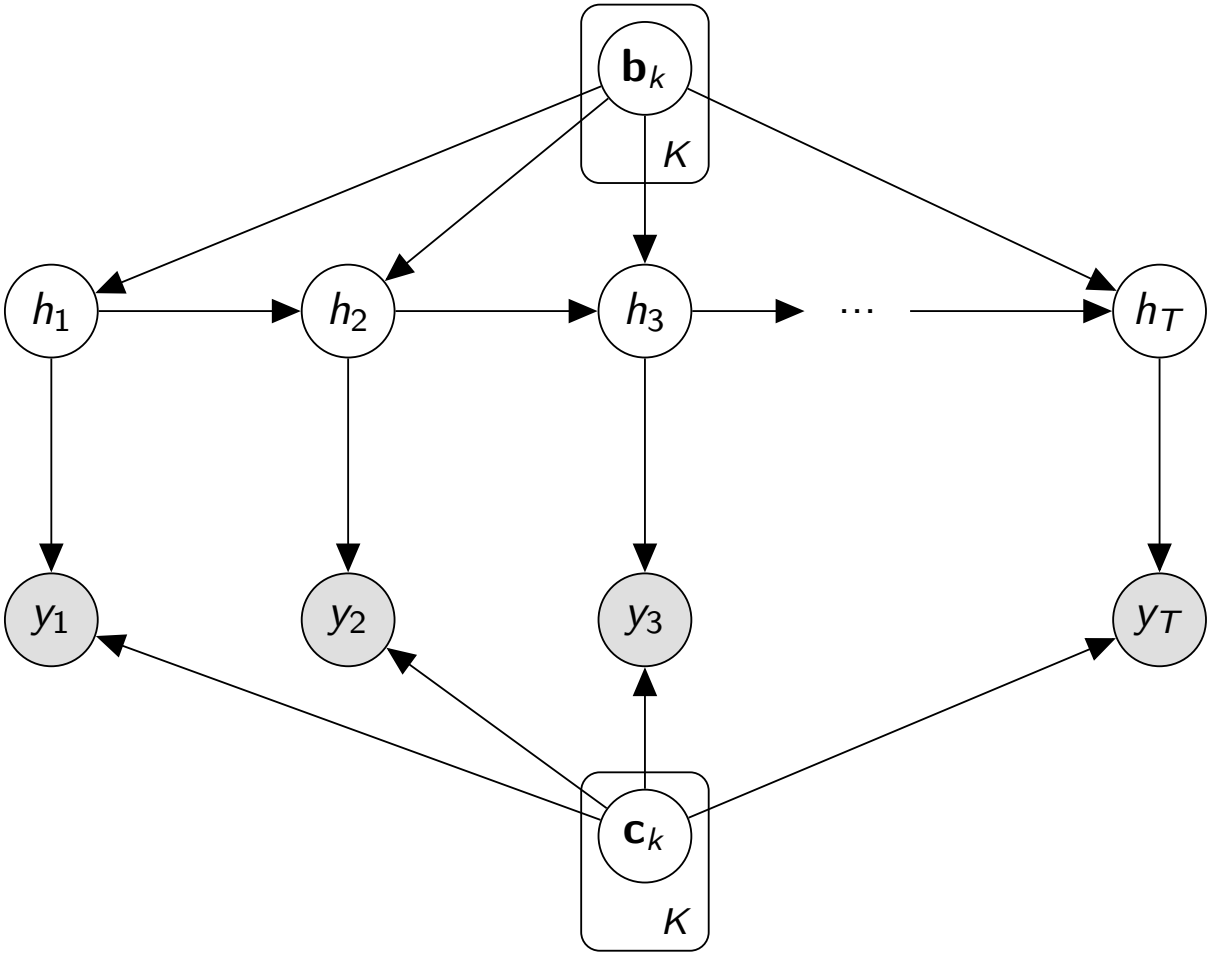
- Putting everything together, the **generative process** becomes:
  1. For each possible hidden state value $k \in \{1, \ldots, K\}$
     a. Draw transition probabilities $\mathbf{b}_k \sim \text{Dirichlet}(\mathbf{b}_k|\boldsymbol{\alpha})$
     b. Draw emission probabilities $\mathbf{c}_k \sim \text{Dirichlet}(\mathbf{c}_k|\gamma)$
  2. For each time $t \in \{1, \ldots, T\}$
     a. Draw new hidden state $h_t \sim \text{Categorical}(h_t|\mathbf{b}_{h_{t-1}})$
     b. Draw observation $y_t \sim \text{Categorical}(y_t|\mathbf{c}_{h_t})$

sample → [0.7 0.1 0.2]

→ sums to one

Hidden Markov models (HMMs)

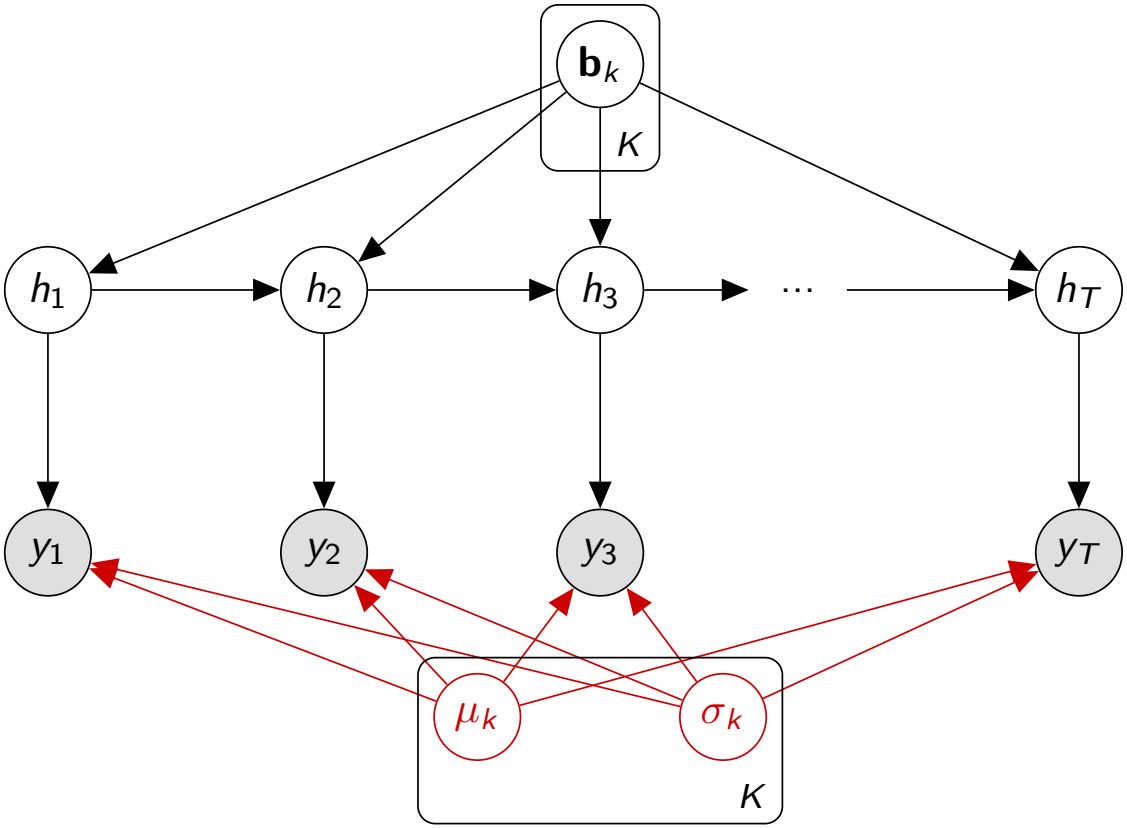
- Alternatively, we can represent the graphical model of the HMM as:



- Both ways are correct... This is one is more similar to the generative process

Extensions: HMMs with continuous observations

- We can **mix discrete** hidden states with **continuous** observations



- Hidden state value h_t defines from which of K Gaussians the observation y_t comes from

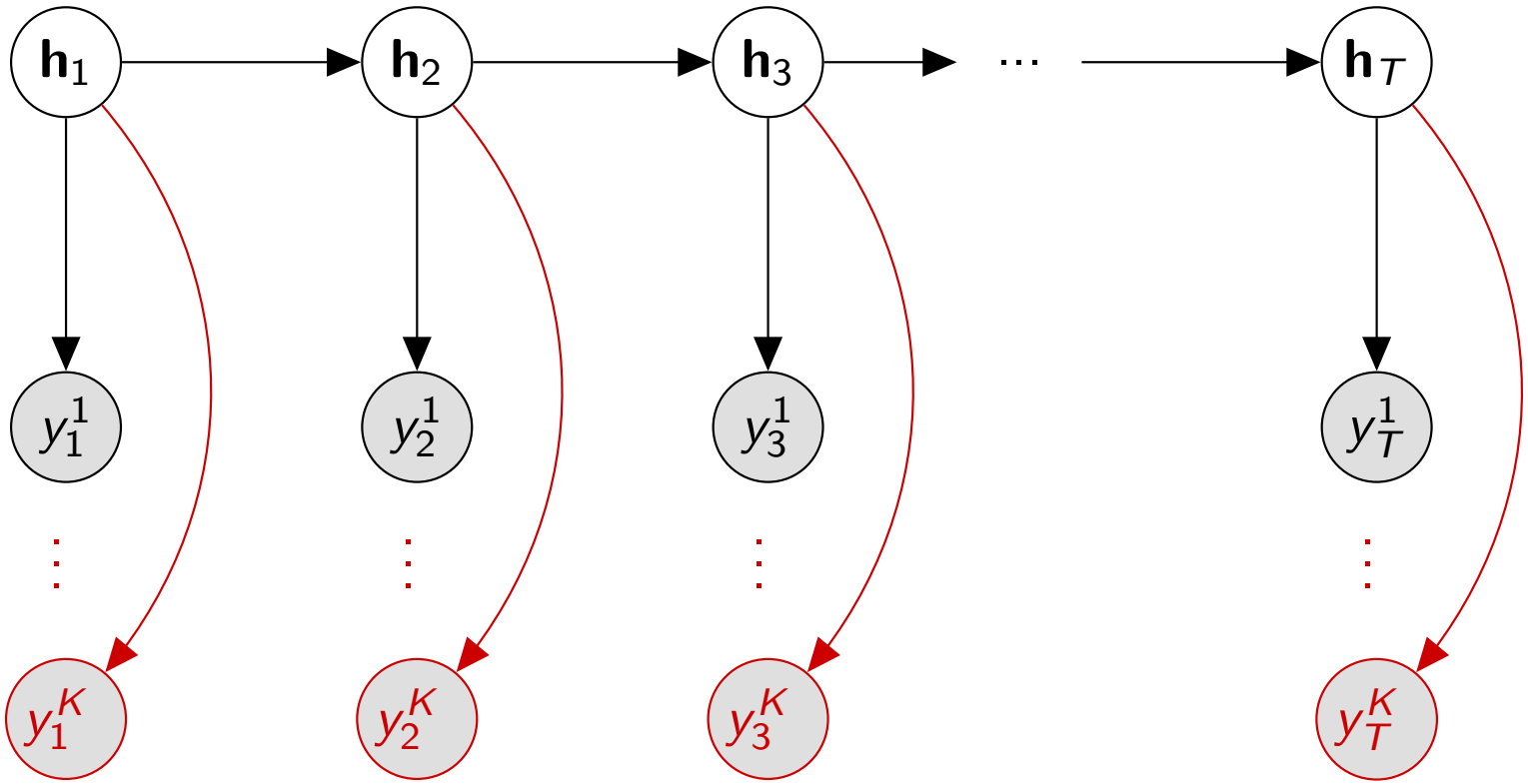
$$y_t \sim \mathcal{N}(y_t | \mu_{h_t}, \sigma_{h_t})$$

- Popular for modeling speech: y_t are sounds (continuous) and h_t are words

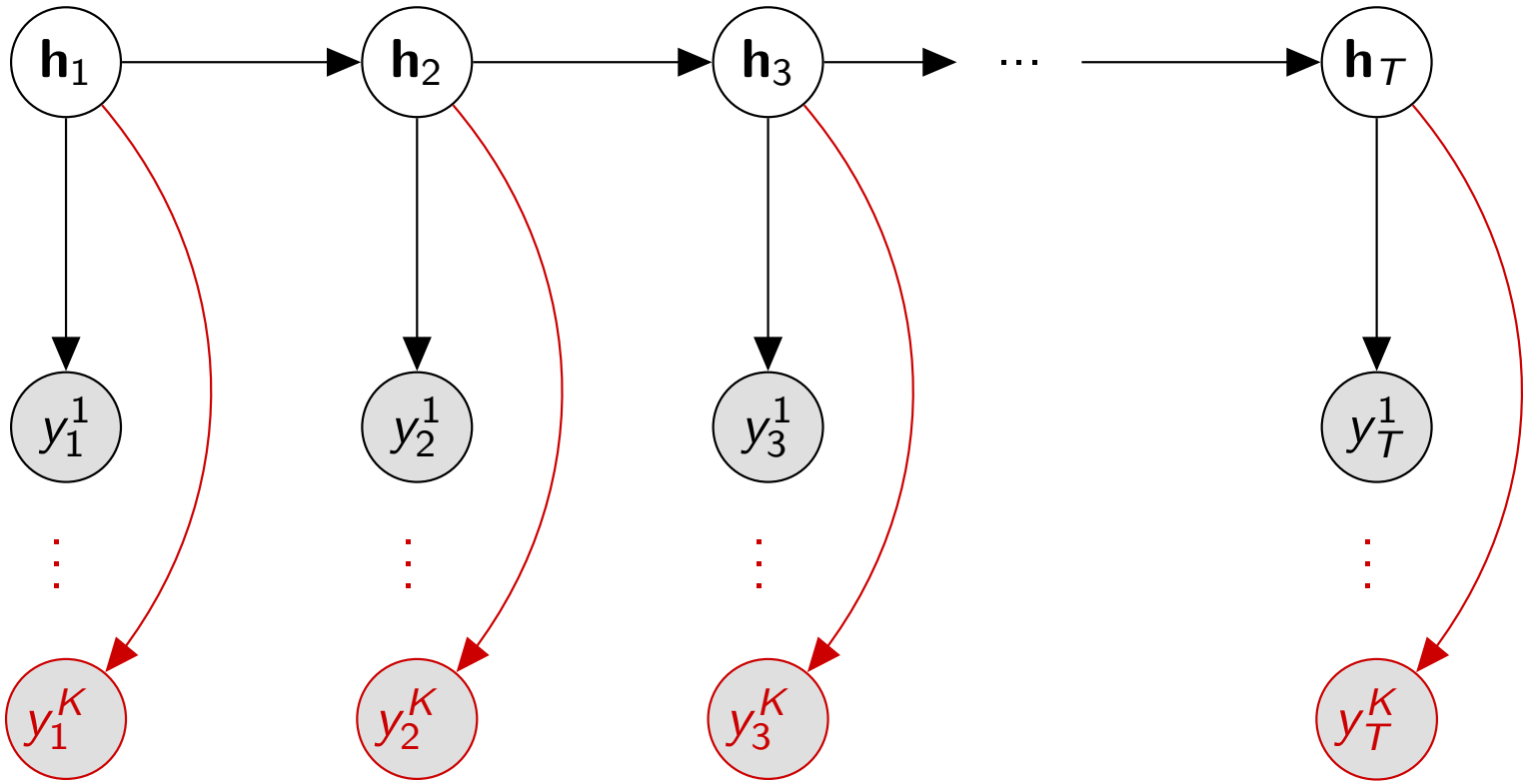
Multivariate state-space models

- What if, rather than a single time-series, we observed multiple time-series? (e.g. multiple sensors)
- Typically, multivariate time-series can exhibit **strong correlations** between the different component series
  - E.g., in our case study, occupancy rates at nearby segments of the freeway should be highly correlated
- Multivariate state-space models allow us to model these correlations

Multivariate state-space models



Multivariate state-space models



- We can group all the K observed outputs at time t in a single vector $\mathbf{y}_t = (y_t^1, \ldots, y_t^K)$
- Model becomes:

$$\mathbf{h}_t \sim \mathcal{N}(\mathbf{h}_t | \mathbf{B}\mathbf{h}_{t-1}, \mathbf{R})$$
$$\mathbf{y}_t \sim \mathcal{N}(\mathbf{y}_t | \mathbf{C}\mathbf{h}_t, \mathbf{\Sigma})$$

model covariances between different time series

Multivariate state-space models

- A particularly important aspect is how we define the covariance matrices $\mathbf{R}$ and $\boldsymbol{\Sigma}$
- Consider a vector $\mathbf{y}$ of size N distributed as a multivariate Gaussian:

$$\mathbf{y} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$$

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- Consider a vector $\mathbf{y}$ of size N distributed as a multivariate Gaussian:

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  - A full covariance matrix $\mathbf{\Sigma}$ also models covariance between the different elements of $\mathbf{y}$ ($N \times N$ parameters! But there are ways to reduce this...)

Multivariate state-space models

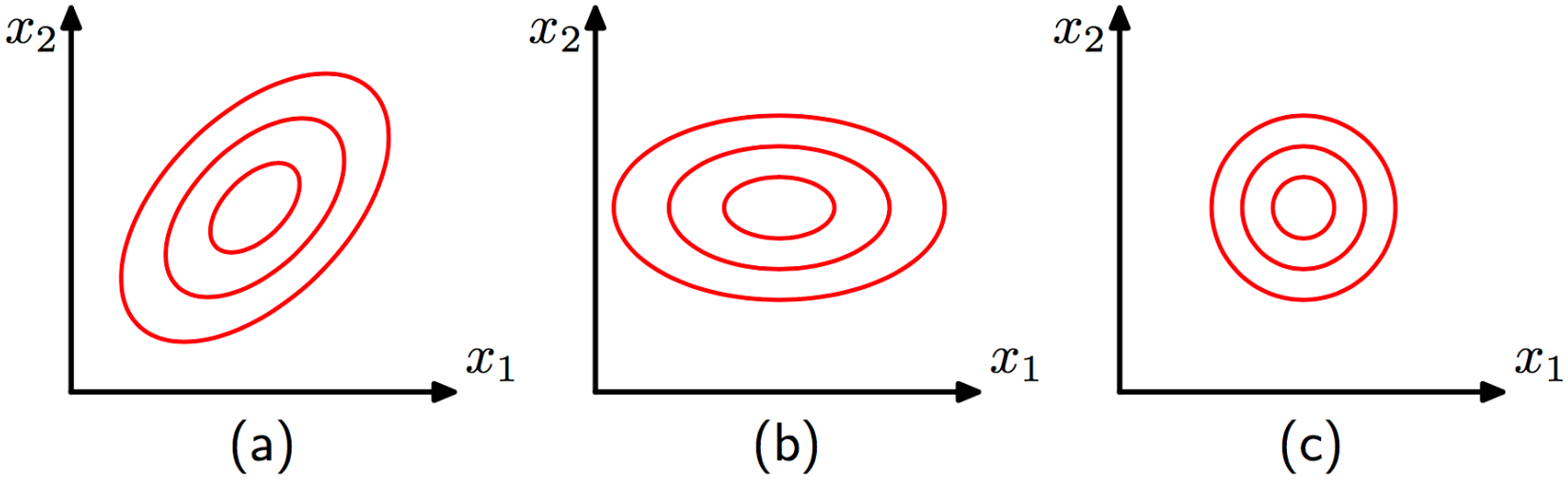

Figure: Bishop, C., 2006. Pattern Recognition and Machine Learning.

(a) A full covariance matrix $\mathbf{\Sigma}$ (models covariances!)

(b) A diagonal covariance of the form $\mathbf{\Sigma} = \text{diag}(\boldsymbol{\sigma})$

(c) An isotropic covariance of the form $\mathbf{\Sigma} = \sigma \mathbf{I}$

Playtime!

- **Multivariate** LDS model of freeway occupancy rates: multivariate forecasting
- See “06 - Temporal models - Part 3.ipynb" notebook